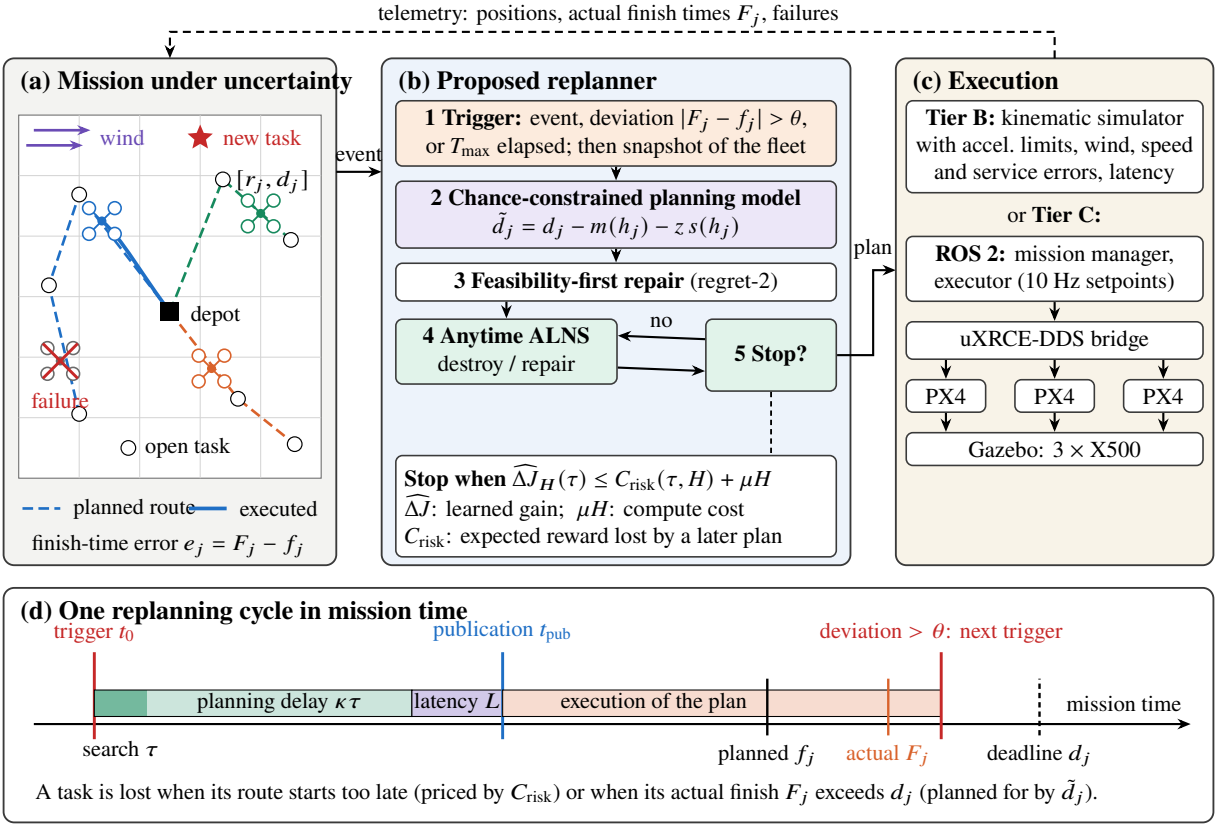

telemetry: positions, actual finish times F_j, failures
(a) Mission under uncertainty
wind
new task
$[r_j, d_j]$
depot
failure
open task
planned route
executed
finish-time error $e_j = F_j - f_j$
event
(b) Proposed replanner
1 Trigger: event, deviation $|F_j - f_j| > \theta$, or $T_{\max}$ elapsed; then snapshot of the fleet
2 Chance-constrained planning model
$\tilde{d}_j = d_j - m(h_j) - z\,s(h_j)$
3 Feasibility-first repair (regret-2)
4 Anytime ALNS
destroy / repair
no
5 Stop?
Stop when $\widehat{\Delta J}_H(\tau) \le C_{\text{risk}}(\tau, H) + \mu H$
$\widehat{\Delta J}$: learned gain; μH: compute cost
C_{risk}: expected reward lost by a later plan
plan
(c) Execution
Tier B: kinematic simulator with accel. limits, wind, speed and service errors, latency
or Tier C:
ROS 2: mission manager, executor (10 Hz setpoints)
uXRCE-DDS bridge
PX4
PX4
PX4
Gazebo: 3 × X500
(d) One replanning cycle in mission time
trigger t_0
publication t_{pub}
deviation > θ: next trigger
planning delay $\kappa\tau$
latency L
execution of the plan
mission time
search τ
planned f_j
actual F_j
deadline d_j
A task is lost when its route starts too late (priced by C_{risk}) or when its actual finish F_j exceeds d_j (planned for by $\tilde{d}_j$).